ALTAIR
QUALITY. VALUE. INTEGRITY.

## Quotation No : Q25000134

Date : 27-Mar-2025

Sudarshan Chemical Industries Limited
CUDDALORE SIPCOTKUDIKADU SIPCOT
Cuddalore
TAMILNADU
607005

## Kind Attention : Mr./Ms. VELMURUGAN

Dear Sir,

Our team understands that selecting gas generation systems is a difficult task.
We have the expertise to help you being a prime manufacturer and supplier of gas generation systems.

Absstem has the resources, facilities, technology, and expertise required to ensure a successful project.
Our approach is low-risk because of our previous experience with large and complex projects.
Post-installation, our shield service division, with diligent accuracy and functionality, oversees a team of technicians who provide warranty and post-warranty support to your newly-installed gas generation systems.
Our customers have relied on scale up their production.

Our products have consistently delivered absolute purity requirements.
The industry's lowest ever production downtime.
Reduce OPEX by at least 70%.

We wish to form a long-term relationship with the goal to develop the most cost-effective and comprehensive solutions for your needs.
With an experienced team of professional personnel already in place, our team is prepared to move swiftly to ensure that we meet critical objectives & timelines.
Please feel free to contact in case of any query.

Looking forward for a positive response.

Thanks and Regards,

## For ALTAIR PNEUMATICS

K. Suresh Babu
Head - Operations
9566198947
www.altairpneumatics.com




nGEN POWERED BY ABSSTEM INDUSTRIAL

# Industrial

# PSA Nitrogen Generator

## Unique Features of Absstem Nitrogen Generator nGEN SH

a. **Lowest Air to Gas ratio** in industry thereby guaranteeing **lowest power consumption**.
b. **10 years*** warranty on Molecular Sieves **(First In Industry)**.
c. Assured Performance for consistent Flow, Pressure and Purity of Nitrogen throughout Product Lifecycle.
d. IIOT ready machine having 100% operator free design for 24X7 operation.
e. Idle mode with auto start/stop features ensuring power savings during low/no demand.
f. Inbuilt dew point transmitter to monitor input compressed air quality as standard scope of supply.
g. Plug n Play machine with an ergonomic design suitable for accessibility.
h. All the Instruments are specially manufactured to make them suitable for Nitrogen gas.
i. All critical performance parameters are being monitored through HMI.
j. First in industry **Comprehensive Maintenance Contract (CMC)** service under Absstem Premium Shield program providing complete peace of mind to customer.

## Ambient Design Parameters

| | | |
|---|---|---|
| a. | Temperature | 0 to 45 Deg. Celsius |
| b. | Relative Humidity | up to 80 % (Max.) |
| c. | Altitude | up to 1000 m above Sea level |

**Absstem Industrial Solutions Pvt. Ltd.**
Plot No. 591, Sector 8, Phase IV
IMT Manesar, Gurugram, Haryana - 122051
Contact - 1800 3010 3394 Email - indsales@absstem.com
CIN: U29256HR2022PTC103387

## Product Description

| | |
|---|---|
| a. Product Name | High Purity PSA Nitrogen Gas Generator |
| b. PSA Module | **nGEN SH** |
| c. Purity Range | 99% - 99.95% |

## General PSA Generation System Layout

A typical Industrial Nitrogen Generation system consists of three parts:

- Compressed Air System: Air Compressor, Pre-filter, Air Dryer, Dry Air Vessel
- PSA $N_2$ Generator
- Nitrogen Storage Tank




| | |
|---|---|
| **Air Compressor** | An Air compressor is used to compress the atmospheric air to a pressure required for the separation of **Nitrogen** from the air. |
| **Refrigeration dryer** | Compressed air then passes through a refrigerated dryer to remove moisture (which is harmful to sieves) from the air. |
| **Air Tank** | Dry air is collected in an air tank to remove the pulsations which serve the dynamic (changing) requirement of air by the PSA system. |
| **Multistage Filtration system** | Before air enters the PSA system, it is passed through multistage filtration removing moisture, particulates and oil content to the specified levels. |
| **PSA system** | Pressurized air is passed through PSA Towers filled with Molecular sieves to separate **Nitrogen** from Air. |
| **Storage Tank** | Industrial Grade **Nitrogen** produced from the PSA system is stored in this tank for continuous supply. |
| **Oxygen Analyser** | Nitrogen produced is monitored by an Oxygen analyser. In case, $O_2$ impurity goes above the specified level, it gives an alarm and switches the plant to vent mode. |



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## HD Touch Screen Display

(User-friendly touch screen interface for easy operation and maintenance)

Absstem industrial Nitrogen generators complies with all industry standards and regulation

- Alarm management system
- Process data display
- Consumables running hours' display
- Data logging via USB drive**
- Graphical representation**




## Additional upgrades

- Digital flow meter with totalizer
- Nitrogen Pressure monitoring for High pressure booster driven plants
- Cylinder manifold and alarm management system
- Remote monitoring
- Nitrogen gas Dew-point monitoring

## International Quality standards

A strict quality procedure, following the leading international pharmacopoeia and Industrial directives, is followed in the selection of all mechanical and electrical instruments, vessel fabrication and designing of Industrial Gas generators.

TÜV NORD ISO 9001 · TÜV NORD ISO 14001 · TÜV NORD ISO 13485 · RIR CERTIFICATION Partners in Growth



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# Absstem Shield Premium

## (Comprehensive Annual Maintenance Contract)




Any Industry is now heavily focused on increased OEE (Overall Equipment Efficiency) and reduced LCC (Life Cycle Cost) while deciding to invest in any equipment. At Absstem, we strive for an everlasting engagement with our customers by not selling them just a product but also provide a long term after-sales support.

We understand that to deliver a consistent performance, periodic health check and timely maintenance of the Nitrogen Generator is extremely important. With Absstem Shield Premium, you invest in complete peace of mind for all maintenance needs pertaining to N2 generators and associated integrated equipment like air compressor, Air dryer etc.

Few benefits of Absstem Shield Premium are listed as below: -

1) Guaranteed assurance of plant performance.
2) Genuine spare parts and consumables being used to provide longevity and durability.
3) Regular visits by our technicians for health monitoring and carrying out the scheduled maintenance services.
4) 24 x 7 On-Call service support to our customers for any basic trouble shooting
5) Emergency visit to attend any unplanned breakdown to any location within 48 hours.
6) No additional surprise charges as all breakdown spares and consumables covered.
7) The contract can be signed at the time of equipment ordering itself and can be renewed on year on year basis.



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# Product Description

| Product | Description | Make | Qty |
|---|---|---|---|
| nGen Nitrogen Generator High Purity | nGEN080SH<br>Nitrogen | | 1 |

| Caption | Specification |
|---|---|
| Gas Generated | Nitrogen |
| Gas Pressure | upto 6.5 Barg |
| FLOW AT 99% PURITY | 205 Nm3/Hr DRY AIR REQUIRED 265 CFM |
| FLOW AT 99.5% PURITY | 171 Nm3/Hr DRY AIR REQUIRED 257 CFM |
| FLOW AT 99.9% PURITY | 127 Nm3/Hr DRY AIR REQUIRED 241 CFM |
| FLOW AT 99.95% PURITY | 114 Nm3/Hr DRY AIR REQUIRED 252 CFM |
| COMPRESSED AIR PRESSURE REQUIRED | 7.5 Barg |
| RECOMMENDED NITROGEN STORAGE | 3000 Liters |
| POWER SUPPLY | 230 V/ 110 V 50/60 Hz Single Phase |
| RECOMMENDED AIR RECEIVER | 2500 Litres Part of Air System |
| INPUT AIR DEW POINT MONITOR | INCLUDED |
| CMS Warranty | 10 Years* |

| Product | Description | Make | Qty |
|---|---|---|---|
| nGen Nitrogen Generator High Purity | nGEN070SH<br>Nitrogen | | 1 |

| Caption | Specification |
|---|---|
| Gas Generated | Nitrogen |
| Gas Pressure | upto 6.5 Barg |
| FLOW AT 99% PURITY | 173 Nm3/Hr DRY AIR REQUIRED 224 CFM |
| FLOW AT 99.5% PURITY | 145 Nm3/Hr DRY AIR REQUIRED 218 CFM |
| FLOW AT 99.9% PURITY | 107 Nm3/Hr DRY AIR REQUIRED 203 CFM |
| FLOW AT 99.95% PURITY | 97 Nm3/Hr DRY AIR REQUIRED 214 CFM |
| COMPRESSED AIR PRESSURE REQUIRED | 7.5 Barg |
| RECOMMENDED NITROGEN STORAGE | 2500 Liters |
| POWER SUPPLY | 230 V/ 110 V 50/60 Hz Single Phase |
| RECOMMENDED AIR RECEIVER | 2000 Litres Part of Air System |
| INPUT AIR DEW POINT MONITOR | INCLUDED |
| CMS Warranty | 10 Years* |

FACTORY ADDRESS
Altair Pneumatics
No. 3/3, (3/142), Ramapuram Road, Sathya Nagar, Manapakkam, Chennai - 600 089.
OFFICE ADDRESS
No. 5/848, 2nd Floor, Aradhana Castle, 5th Block, Mogappair West, Chennai - 600 037.
info@thealtair.in | www.thealtair.in | +91 44 4558 3550

# Commercial Offer

| S No | Item Code | Item Name | Description | HSN | Qty | Unit Price | GST(%) | Amount |
|---|---|---|---|---|---|---|---|---|
| 1 | nGEN080SH | nGEN080SH | nGen Nitrogen Generator High Purity | 840510 | 1 | 48,37,200.00 | 18.00 | 48,37,200.00 |
| 2 | nGEN070SH | nGEN070SH | nGen Nitrogen Generator High Purity | 840510 | 1 | 43,22,400.00 | 18.00 | 43,22,400.00 |

| | |
|---|---|
| Sub Total | 91,59,600.00 |
| GST(%) @18.00 | 16,48,728.00 |
| Grand Total | 1,08,08,328.00 |

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# Terms And Conditions

| | | |
|---|---|---|
| 1. | EXCLUSIONS & CLIENTS SCOPE | All erection / civil work, All Power Cables & Cabling out of battery limit of PSA Plant, Cable Trenches etc. Unloading / Shifting of Gas Plant at Clients site will be in Client Scope. Additional gas piping, cooling water (if required). |
| 2. | SUPERVISION OF ERECTION & COMMISSIONING | There shall be no commissioning charges, however you shall arrange freeTo & Fro Air Ticket, lodging / boarding, and local conveyance for our Engineers during supervision of Installation of Gas Plant at site on free of cost basis. Our Engineers will be there for Supervision only. Any manpower / fitter required for installation work shall be provided by you. |
| 3. | P&F | 3% Extra. |
| 4. | Freight | Extra at Actuals. |
| 5. | Insurance | In your Scope. |
| 6. | Payment Terms | 50% Advance with your Prestegious Purchase Order and Balance against PI. |
| 7. | Delivery | 6-8 Weeks from the PO and Advance. Rest can be confirmed for Availability. |
| 8. | Warranty | We hereby confirm that, our equipment shall have Suppliers warranty for a period of 12 months from the date of Commissioning or 18 months from date of Dispatch whichever is earlier. |
| 9. | GST | As per the Commercial Table. |
| 10. | Remarks | . |

Hope you will find our offer in line with your requirement.

Should you require any further clarification/information, please feel free to contact us.

Thanking you, and assuring you of our best attention at all times.

Very Yours truly,

For ALTAIR PNEUMATICS

K. Suresh Babu

Head - Operations

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